PROVISIONAL PATENT APPLICATION

Hull-Integrated Boundary Translation and Exterior-Custody Control Apparatus and Methods

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This specification is complete and self-contained. Companion applications are nonessential background only. Every example under PROPHETIC EXAMPLES is prospective; no force-free translation, anomalous displacement, qualified boundary-translation map, or flight-capable vehicle is represented as achieved.

Portfolio prophetic-example statement: All examples identified as prophetic are prospective; no prophetic example is represented as an experiment performed or a result achieved.

CROSS-REFERENCE TO RELATED APPLICATIONS

This application belongs to a coordinated series concerning controlled angular-momentum coherence and boundary engineering. A companion transportation application concerns vehicles driven by calibrated support-force modules. The present application is distinct: it treats the vehicle hull or distributed boundary itself as the translation actuator, preserves an interior custody set, selects an exterior custody/address state, and distinguishes boundary translation from propulsion by ordinary through-vehicle impulse.

FIELD

Vehicle boundaries; distributed resonant and phase-controlled hull sectors; exterior address and custody control; translation-response calibration; occupant and payload invariant preservation; propulsion metrology; whole-platform energy and momentum accounting; and safe operation of test articles and vehicles.

BACKGROUND

1. Conventional propulsion changes vehicle motion by exchanging momentum through propellant, aerodynamic surfaces, contact forces, radiation, interaction with an external field or prepared guideway, or another measurable reaction channel.
2. Distributed arrays, active skins, phased apertures, structural health systems, metamaterial surfaces, and vehicle-wide control networks provide manufacturable boundary hardware. They do not by themselves teach a hull that controls the vehicle's exterior custody relation while leaving the interior in a locally low-load state.

3. Reports of unusual vehicles often combine displacement, low apparent occupant acceleration, broad hull emission, and radar visibility. Such reports do not establish a mechanism. They motivate a discriminating apparatus that measures the complete conjunction and can return an ordinary-propulsion classification or a null.
4. The applicant's architecture divides the problem into an interior custody book and an exterior posting book. A hull-sector controller maintains interior invariants while commanding a distributed boundary state associated with an exterior translation setpoint. A qualification fixture separately measures vehicle displacement, local specific force, support and suspension forces, radiated momentum, expelled mass, aerodynamic and acoustic loads, magnetic and electrostatic forces, thermal recoil, ground coupling, and every actuator reaction.
5. The first realizable rung is an instrumented boundary-sector or suspended test article that produces a translation-response map, including a null map. Scale-up to a vehicle is conditional on a repeatable residual not accounted for by conventional impulse and on closure of the full energy and momentum ledger.

SUMMARY

A boundary-translation apparatus (100; FIG. 1) comprises a body or vehicle structure, plural controllable hull sectors, a phase and timing distribution network, an interior custody registry, exterior address sensors, a translation controller, complete reaction witnesses, a calibration memory, and a safe-neutral state. The controller receives an exterior translation setpoint, selects a boundary state from an empirically qualified map, preserves stored interior limits, commands the hull sectors, and reports achieved displacement, velocity, acceleration, local specific force, reaction impulse, and uncertainty.

Qualification (FIGS. 6–8) compares active states with phase-scrambled, spatially scrambled, absorbed-energy-matched, reaction-injected, conventional-thruster, dummy-load, enclosure, orientation, and support-swap controls. A claimed direct boundary-translation result requires displacement or changed exterior motion not accounted for by measured ordinary impulse, while occupant-local specific force and interior strain remain within the declared custody envelope.

Products include a boundary-sector test article, a calibrated translation module, a plural-sector hull, a suspended or free-fall qualification platform, and a prophetic occupied vehicle.

BRIEF DESCRIPTION OF THE DRAWINGS

- FIG. 1 — Hull-integrated boundary-translation apparatus and separate interior, boundary, exterior, and support books.
- FIG. 2 — Distributed hull sector with resonant element, phase trim, local witnesses, reaction mount, and data interface.
- FIG. 3 — Translation controller state machine from map validation through command, custody monitoring, and safe neutral.

- FIG. 4 — Interior custody invariants and exterior translation setpoint.
- FIG. 5 — Whole-platform energy, momentum, angular-momentum, thermal, radiative, acoustic, expelled-mass, and support ledger.
- FIG. 6 — Suspended reference test article and independently instrumented source/reaction frame.
- FIG. 7 — Free-flight or free-fall qualification range with independent tracking and environmental witnesses.
- FIG. 8 — Randomized control matrix and acceptance gates L1–L10.
- FIG. 9 — Empirical translation-response map, permissible interpolation region, and array allocation.
- FIG. 10 — Prophetic vehicle using a plural-sector hull, interior custody controller, and graceful-degradation modes.
- FIG. 11 — Qualification-to-deployment, provenance binding, validity states, and recalibration.

DEFINITIONS

1. **Boundary sector** — an addressable portion of a body or vehicle boundary having one or more controllable resonant, phase, polarization, impedance, orientation, thermal, mechanical, electric, magnetic, acoustic, optical, or other boundary states.
2. **Interior custody set** — the declared occupants, payloads, clocks, structural relations, stored records, environmental conditions, and other retained organizations whose invariants are monitored during a translation attempt.
3. **Exterior custody/address state** — the measured relation of the test article to an external reference network, determined by independent tracking, timing, ranging, imaging, inertial, radar, optical, radio-frequency, or other declared observations.
4. **Boundary-state vector** — commanded and measured states of all operated sectors together with phase, amplitude, frequency, polarization, timing, temperature, support, environment, and configuration.
5. **Translation setpoint** — a commanded displacement, velocity, trajectory, exterior address, or time-indexed profile relative to a declared external reference.
6. **Local specific force** — accelerometer or proof-mass reading at an identified interior location, distinguished from kinematic acceleration obtained from external tracking.
7. **Ordinary impulse account** — the integral of measured forces or momentum flux through propellant, air, radiation, cables, bearings, supports, fields, expelled matter, acoustic output, thermal radiation, or other conventional reaction channels.
8. **Boundary-translation residual** — externally measured motion remaining after the ordinary impulse account and uncertainty are applied. An unexplained residual is not a qualified direct-translation result until all gates pass.
9. **Empirical translation-response map** — measured relation between boundary-state vector and displacement, velocity, acceleration, local specific force, strain, reaction impulse, energy, heat,

emissions, and uncertainty, with provenance and operating limits. A map may contain only null or conventional responses.

10. **Qualified boundary state** — a mapped operating point that passes the applicable acceptance gates, including independent confirmation.
11. **Interior-invariant envelope** — stored limits on local specific force, strain, pressure, temperature, clock ratios, orientation, vibration, acoustic level, radiation, power quality, and custody continuity.
12. **Safe neutral** — removal of dynamic sector commands in a stored sequence, retention of passive support required for safety, preservation of logs, and transition to conventional stabilization or containment.
13. **Direct boundary translation** — prophetic operation in which the exterior custody/address state changes without the change being accounted for by ordinary impulse through the test article, while the interior custody set remains inside its invariant envelope.
14. **Conventional response** — motion accounted for by measured ordinary impulse, including useful active-surface or field propulsion.
15. **Validity states** — qualifying, valid, degraded, out-of-envelope, invalid, contained, and safe neutral.

THEORETICAL FOUNDATION AND OPEN TRANSFER QUESTION

The motivating POAMS framework separates internal constitution from exterior custody. The hull is treated not merely as a pressure vessel but as the boundary at which the retained interior is posted into an exterior address record. The conjecture is that a controlled boundary state may change the exterior posting while preserving the internal closure complex.

The framework does not supply a validated material law. The open empirical relation is

Y = R_B(x_B, C_I, X_E, t),

where **x_B** is the boundary-state vector, **C_I** the interior custody state, **X_E** the external environment and reference configuration, and **Y** the complete motion, specific-force, reaction, thermal, emission, and support record. **R_B** may contain only conventional effects or zero.

The discriminating quantity is not displacement alone but the conjunction

{external kinematics, interior specific force, structural strain, ordinary impulse, support ledger}.

PRODUCT ARCHITECTURE (FIG. 1)

The test article or vehicle structure (110) carries plural boundary sectors (120). Each sector has a local drive and phase trim (122), local state sensors (124), and an instrumented mechanical/electrical interface (126). A common reference distributor (130) supplies timing and permitted phase relations while recording actual delivered phase.

The interior custody controller (140) monitors the invariant envelope. Exterior metrology (150) uses independent references not mechanically or computationally dependent on the hull controller. The complete reaction witness set (160) measures supports, feeds, emitted momentum, expelled matter, air and acoustic loading, magnetic/electric interaction, thermal radiation, and ground motion. Calibration memory (170) holds the map and provenance. Translation controller (180) accepts a setpoint only in a valid state. Safe-neutral subsystem (190) returns the system to a bounded condition.

DEPLOYED METHOD (FIG. 3)

The controller verifies map provenance, executes a neutral baseline, checks external tracking independence, receives a translation setpoint, confirms that the requested state lies in the permissible region, computes a sector allocation from a measured array map, commands bounded ramps, monitors interior and reaction witnesses, reports achieved motion with uncertainty and reaction classification, and enters safe neutral on any limit violation. Direct-translation mode is disabled unless the map contains an independently confirmed boundary-translation residual at the requested scale and configuration.

LAYERED SCOPE

- **L0:** raw boundary, kinematic, specific-force, strain, reaction, emission, and energy records.
- **L1:** useful conventional active-boundary propulsion or motion control.
- **L2:** unexplained boundary-correlated motion residual.
- **L3:** independently confirmed direct boundary translation within a bounded configuration.
- **L4:** plural-sector occupied vehicle operation within a measured map.

No layer is inferred from a lower layer.

QUALIFICATION APPARATUS AND GATES (FIGS. 6–8)

The reference test article is suspended from a torsion, flexure, air-bearing, magnetic-bearing, pendulum, drop-tower, parabolic-flight, orbital, or other characterized low-force platform. Feeds are mechanically isolated or carried onboard. A second instrumented frame records drive and support reactions. Independent optical and radio-frequency tracking measure external kinematics. The randomized matrix includes drive-off, resonant, detuned, phase-scrambled, spatially scrambled, polarity-reversed, absorbed-energy-matched, dummy-sector, matched-heater, calibrated-force-injection, support-swap, enclosure, and conventional-thruster states.

- **L1 repeatability:** motion and boundary state repeat across blocks and days;
- **L2 tracking independence:** at least two independent external tracking modalities agree;
- **L3 reaction closure:** supports, feeds, radiation, expelled mass, field interaction, air, acoustic, thermal, and ground channels cannot account for a claimed residual;

- **L4 interior custody:** local specific force, strain, clocks, and environmental witnesses remain within the declared envelope;
- **L5 phase and spatial contrast:** response follows held-out phase or sector allocation rather than absorbed energy alone;
- **L6 orientation/environment:** Earth orientation, enclosure, pressure, airflow, magnetic/electric environment, and platform configuration are controlled;
- **L7 conservation:** energy, momentum, and angular-momentum accounts close within uncertainty, with any unmeasured term identified rather than called hidden;
- **L8 statistics:** exploratory search and frozen confirmation are separated and multiplicity controlled;
- **L9 safe interruption:** drive loss, sector fault, and reference loss enter safe neutral without uncontrolled motion; and
- **L10 independent confirmation:** a second article, range, apparatus, or laboratory reproduces the response surface.

EMBODIMENTS

E1—single-sector reaction test. One resonant or phase-controlled panel is mounted on an instrumented reaction frame. The experiment maps ordinary recoil, radiation pressure, acoustic thrust, cable force, thermal drift, and field interaction. This embodiment usefully qualifies the sensor chain even if all motion is conventional.

E2—plural-sector suspended article. Four to 64 sectors are distributed around a rigid shell. The controller compares common-phase, traveling-phase, opposing-phase, phase-scrambled, and sequential allocations. A measured allocation map captures cross-coupling and does not presume additivity.

E3—interior custody capsule. A central independently instrumented capsule records local specific force, six-axis strain, vibration, acoustic level, clocks, temperature, pressure, magnetic/electric fields, and power quality. It is mechanically distinguishable from the sector reaction frame.

E4—conventional active-skin product. If measured operation produces useful motion entirely accounted for by ordinary impulse, the same controller, map, sector hardware, and safety system operate as an active-surface or field-propulsion device. Such utility does not establish direct translation.

E5—prophetic vehicle (FIG. 10). A closed hull carries plural independently addressable sectors, onboard energy, exterior tracking, an interior custody controller, and conventional emergency stabilization. Translation commands are accepted only from a vehicle-scale map. No flight result is represented.

PROPHETIC EXAMPLES

P1—one-meter suspended shell. A shell carrying 24 sectors is tested at sub-newton force sensitivity with onboard power and independent optical/radar tracking. The initial output is a complete ordinary impulse budget and map, including a null.

P2—allocation scaling. One, two, four, eight, and 24 sectors are operated sequentially, coherently, and phase scrambled. Array gain, cancellation, mode splitting, cross-talk, heat, and structural reaction are measured rather than extrapolated.

P3—contained translation progression. Only after L1–L10 confirmation does testing progress from suspended motion to a guided low-energy displacement, then free-flight of an unoccupied instrument package, then an occupied test under separate safety authorization.

POAMS-NATIVE DIRECT-OPERATION EMBODIMENT

In the prophetic native embodiment, the hull maintains the internal closure and custody of the vehicle while selecting a successor exterior address relation. Motion is recorded as controlled reposting of the exterior boundary rather than acceleration of every interior constituent through intervening projected addresses. This interpretation is not required for ordinary operation of the qualification apparatus and is not represented as achieved.

CLAIMS

1. A boundary-translation apparatus comprising: a body or vehicle structure having a plurality of addressable boundary sectors; a drive and phase-distribution system configured to command a boundary-state vector of the sectors; an interior custody controller monitoring an invariant envelope for an interior custody set; independent exterior metrology measuring motion relative to an external reference; reaction witnesses measuring support force, feed force, radiated momentum, expelled mass, environmental interaction, and thermal output; a calibration memory storing an empirical translation-response map with provenance, uncertainty, a permissible region, and safe-neutral parameters; and a translation controller configured to receive a translation setpoint, allocate sector commands from the map, monitor the interior custody controller and reaction witnesses, report achieved motion and ordinary impulse with uncertainty, and enter safe neutral on a validity failure.
2. The apparatus of claim 1, wherein each sector comprises a resonant, phased, polarized, impedance-controlled, mechanical, acoustic, optical, electric, magnetic, or hybrid boundary element with local state sensing.
3. The apparatus of claim 1, wherein the interior invariant envelope includes local specific force, strain, clock ratio, vibration, acoustic level, temperature, pressure, field exposure, and custody continuity.
4. The apparatus of claim 1, wherein the exterior metrology comprises at least two independent modalities selected from optical tracking, radar, radio-frequency ranging, lidar, inertial tracking, and time-transfer ranging.
5. The apparatus of claim 1, wherein the reaction witnesses include a separately loaded source or support frame and the controller classifies motion accounted for by the measured reaction as conventional response.

6. The apparatus of claim 1, wherein the map stores responses for common-phase, traveling-phase, opposing-phase, phase-scrambled, sequential, and inactive-sector states.
7. The apparatus of claim 1, wherein direct-boundary-translation mode is enabled only when the map contains an independently confirmed operating point having an external motion residual not accounted for by the ordinary impulse account.
8. The apparatus of claim 1, wherein safe neutral removes dynamic sector commands in a stored sequence and transfers control to a conventional stabilization or containment system.
9. A method of qualifying a boundary-translation apparatus, comprising operating a test article having plural boundary sectors in randomized drive-off, active, detuned, phase-scrambled, spatially scrambled, polarity-reversed, absorbed-energy-matched, dummy-sector, matched-heater, calibrated-force-injection, support-swap, and conventional-thruster states; measuring external kinematics by independent modalities; measuring interior specific force and strain; measuring support, feed, radiative, expelled-mass, field-interaction, aerodynamic, acoustic, thermal, and ground reaction channels; fitting an ordinary impulse account; estimating a boundary-correlated motion residual with uncertainty; applying predetermined acceptance gates; and writing a translation-response map or null map with provenance and operating limits.
10. The method of claim 9, further comprising separating an exploratory phase and sector scan from a frozen blinded confirmation and controlling for multiple comparisons.
11. The method of claim 9, further comprising operating sector counts in a progression and measuring an array allocation rather than presuming additivity.
12. The method of claim 9, further comprising comparing external kinematic acceleration with local specific force at a plurality of interior locations.
13. The method of claim 9, wherein a response accounted for by ordinary impulse is retained as a calibrated conventional active-boundary propulsion map.
14. The method of claim 9, wherein an unexplained residual is not classified as direct boundary translation until independently confirmed on a second article, apparatus, range, or laboratory.
15. A deployed method comprising validating provenance of a stored translation-response map; executing a neutral baseline; receiving a translation setpoint; determining that the setpoint and environment lie within a permissible region; allocating commands among plural hull sectors from a measured array map; applying the commands while monitoring interior custody and reaction witnesses; reporting achieved displacement, velocity, acceleration, local specific force, ordinary impulse, energy, and uncertainty; and entering safe neutral when a witness exceeds a stored limit.
16. The method of claim 15, wherein the setpoint is a trajectory and the allocation is updated in closed loop from independent exterior tracking while remaining bounded by the map.
17. The method of claim 15, wherein a degraded mode reduces the admitted sector set, speed, or trajectory authority while preserving conventional stabilization.
18. A boundary-sector module comprising a controllable boundary element, a local drive and phase trim, local temperature, phase, amplitude, strain, and reaction sensors, a module identifier, a

calibration memory, and a data interface reporting validity and measured state to a hull controller.

19. A plural-sector hull comprising a plurality of modules of claim 18, an interior custody controller, independent exterior tracking, a system calibration memory holding a measured array allocation map, and a controller that refuses extrapolation outside the map.
20. The hull of claim 19, further comprising onboard energy storage and mechanically isolated conventional emergency stabilization.
21. A prophetic direct-boundary-translation vehicle comprising a closed hull having plural phase-addressable sectors; an interior custody set; an exterior address selector; a controller configured to command a successor exterior address state while preserving declared interior invariants; complete energy and momentum witness systems; and a safe-neutral system, wherein achieved translation magnitude and tariff are outputs of operation and not fixed limitations.
22. A prophetic method of operating the vehicle of claim 21, comprising maintaining an interior closure and custody record, selecting an exterior successor address, commanding a distributed hull boundary state, measuring external displacement and interior specific force, reconciling support and reaction ledgers, and reporting either direct boundary translation, conventional propulsion, an identified artifact, or a null.

ABSTRACT

A hull-integrated boundary-translation apparatus uses plural addressable boundary sectors, phase and timing distribution, independent exterior tracking, an interior custody controller, complete reaction witnesses, and a stored empirical translation-response map. A translation controller allocates sector commands for a requested displacement or trajectory, monitors local specific force, strain, support, feed, radiative, expelled-mass, environmental, and thermal channels, reports achieved motion and ordinary impulse with uncertainty, and enters safe neutral on a validity failure. Randomized phase, spatial, thermal, dummy, reaction-injection, support-swap, and conventional-thruster controls produce a conventional-response map, an artifact classification, a null, or an independently confirmed residual. Boundary-sector test articles, plural-sector hulls, and conventional active-surface embodiments are disclosed. A prophetic vehicle preserves an interior custody set while commanding an exterior address relation; no force-free translation or flight-capable result is represented as achieved.

Hull-integrated boundary-translation apparatus and separate interior, boundary, exterior, and support books

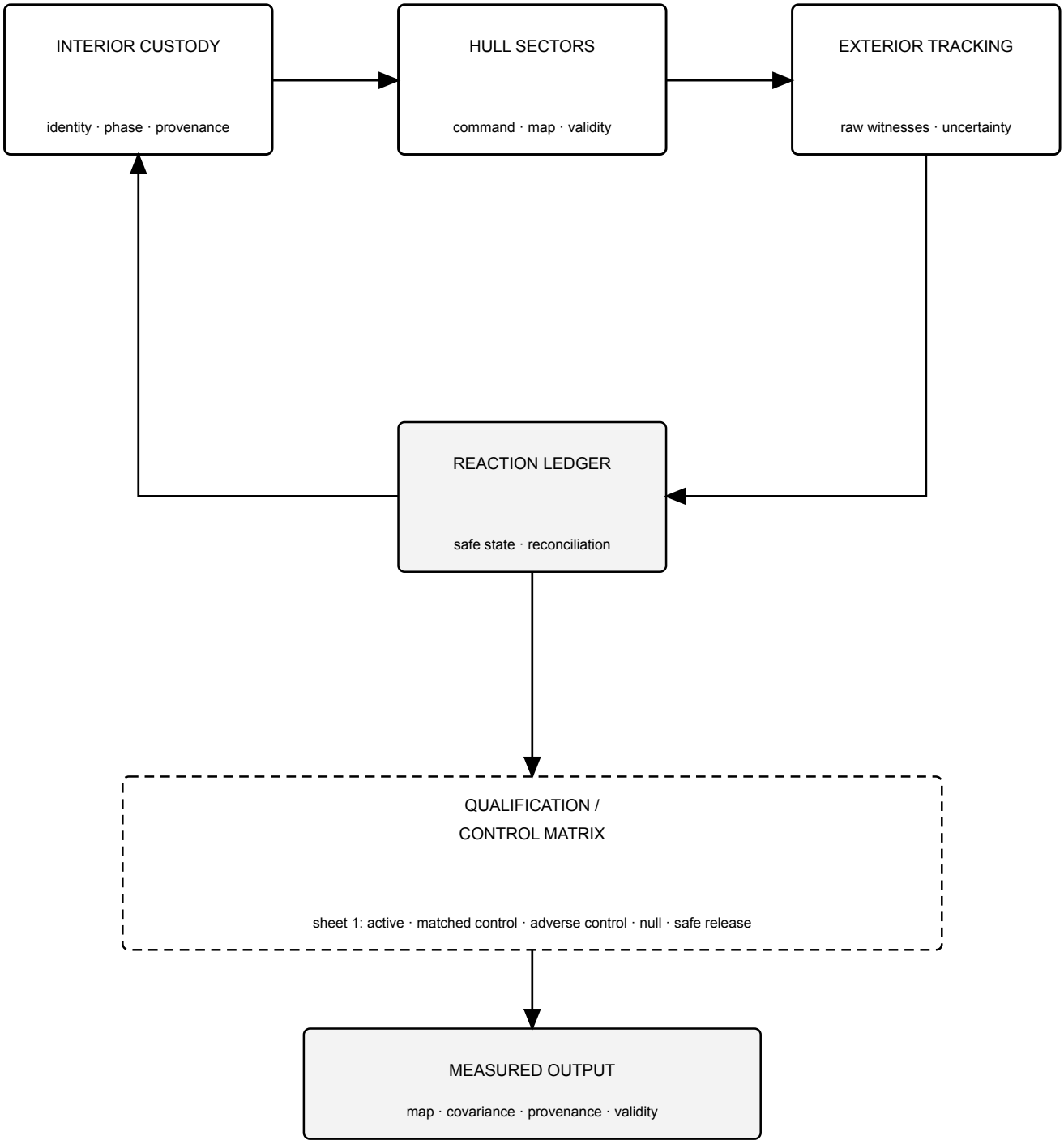


FIG. 1

Hull-integrated boundary-translation apparatus and separate interior, boundary, exterior, and support books.

Distributed hull sector with resonant element, phase trim, local witnesses, reaction mount, and data interface

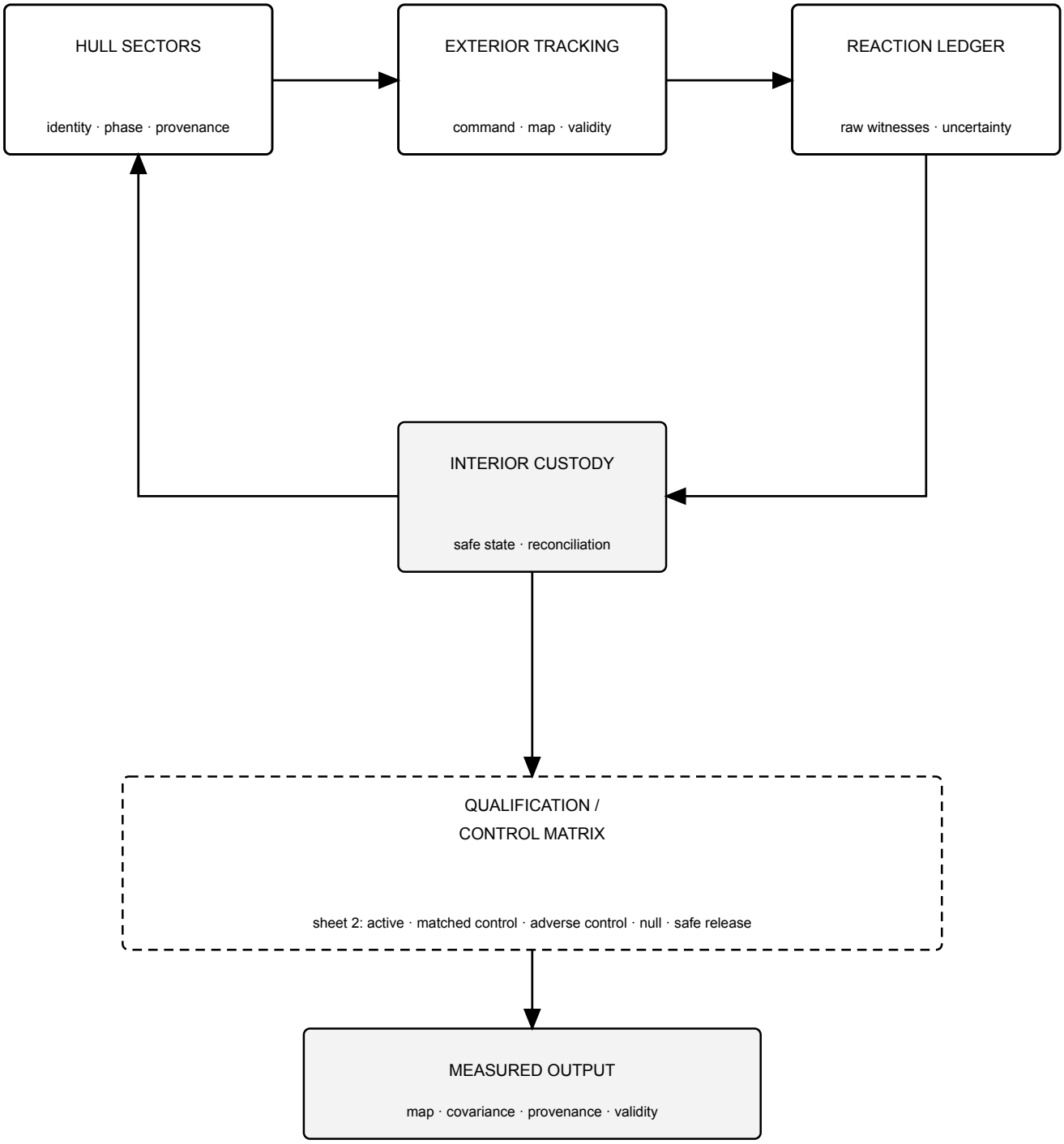


FIG. 2

Distributed hull sector with resonant element, phase trim, local witnesses, reaction mount, and data interface.

Translation controller state machine from map validation through command, custody monitoring, and safe neutral

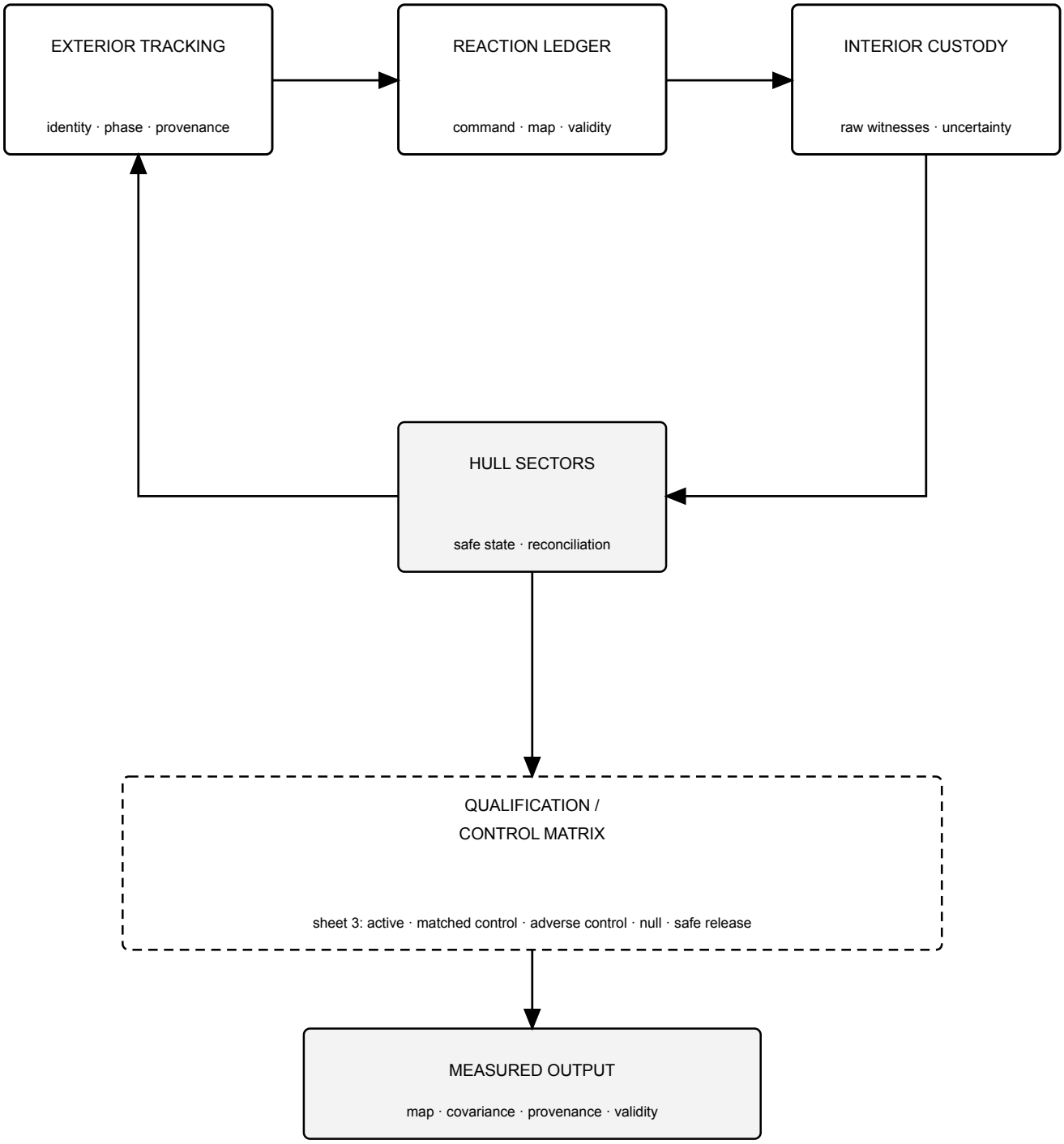


FIG. 3

Translation controller state machine from map validation through command, custody monitoring, and safe neutral.

Interior custody invariants and exterior translation setpoint

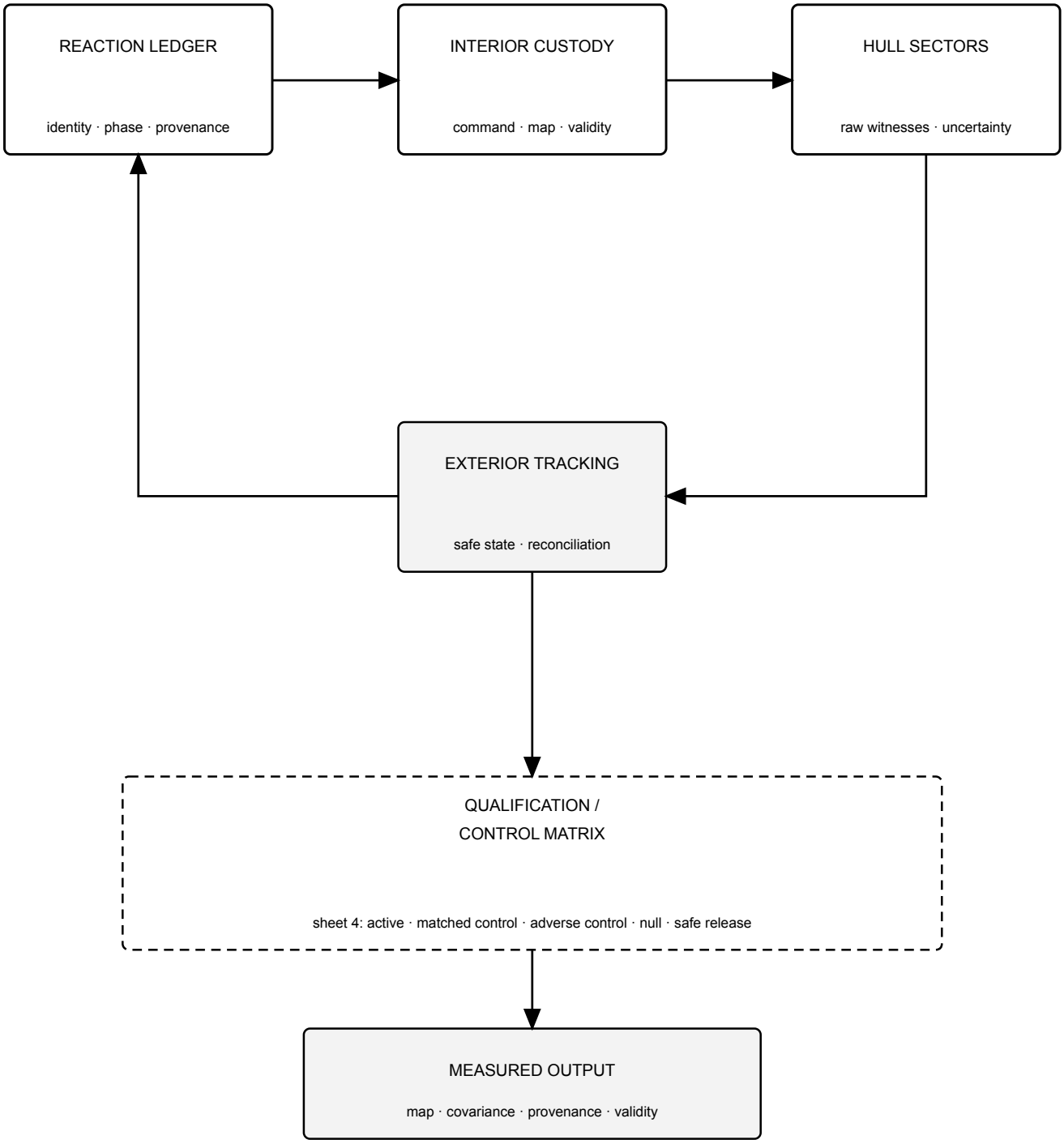


FIG. 4
Interior custody invariants and exterior translation setpoint.

Whole-platform energy, momentum, angular-momentum, thermal, radiative, acoustic, expelled-mass, and support ledger

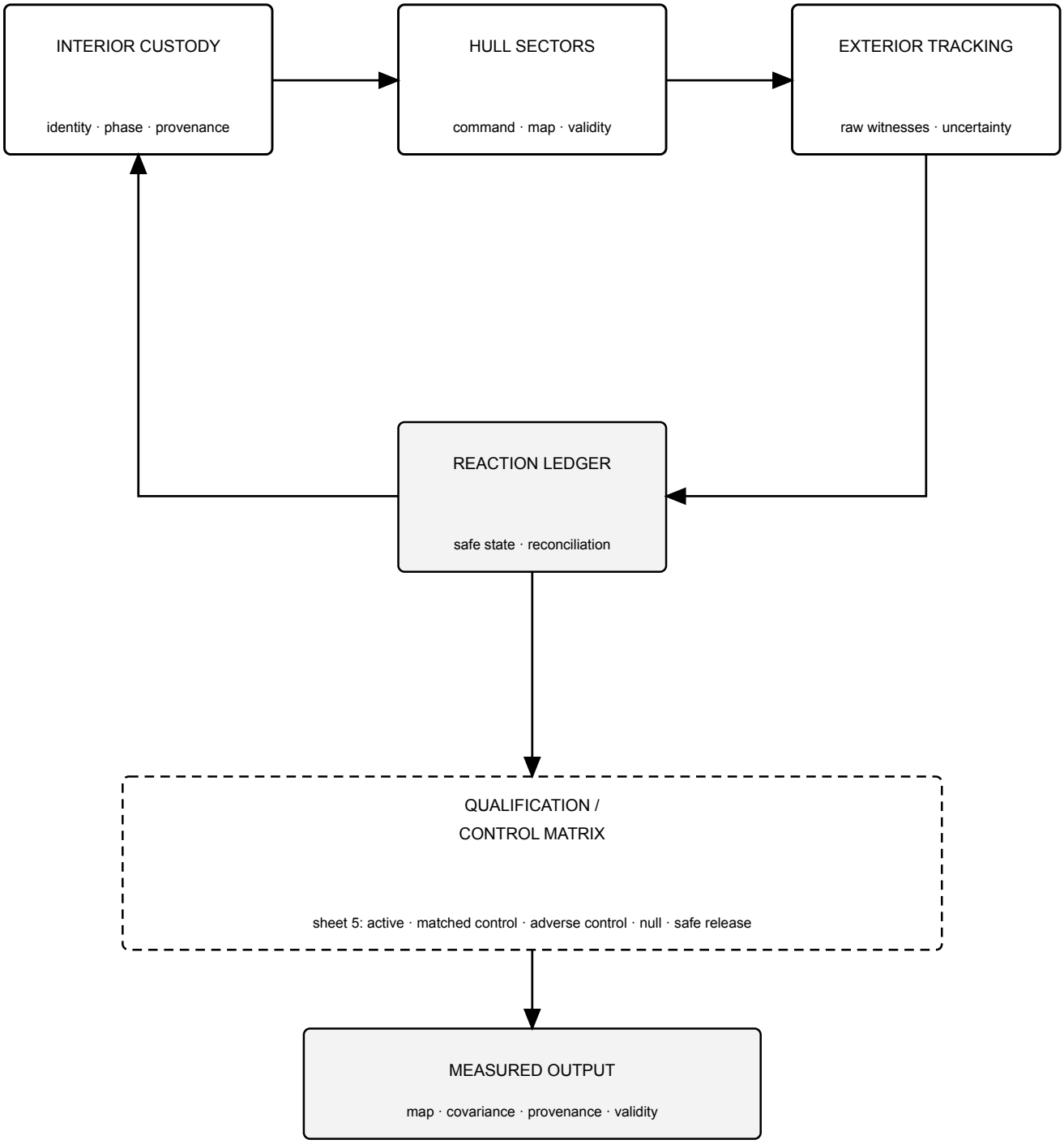


FIG. 5

Whole-platform energy, momentum, angular-momentum, thermal, radiative, acoustic, expelled-mass, and support ledger.

Suspended reference test article and independently instrumented source/reaction frame

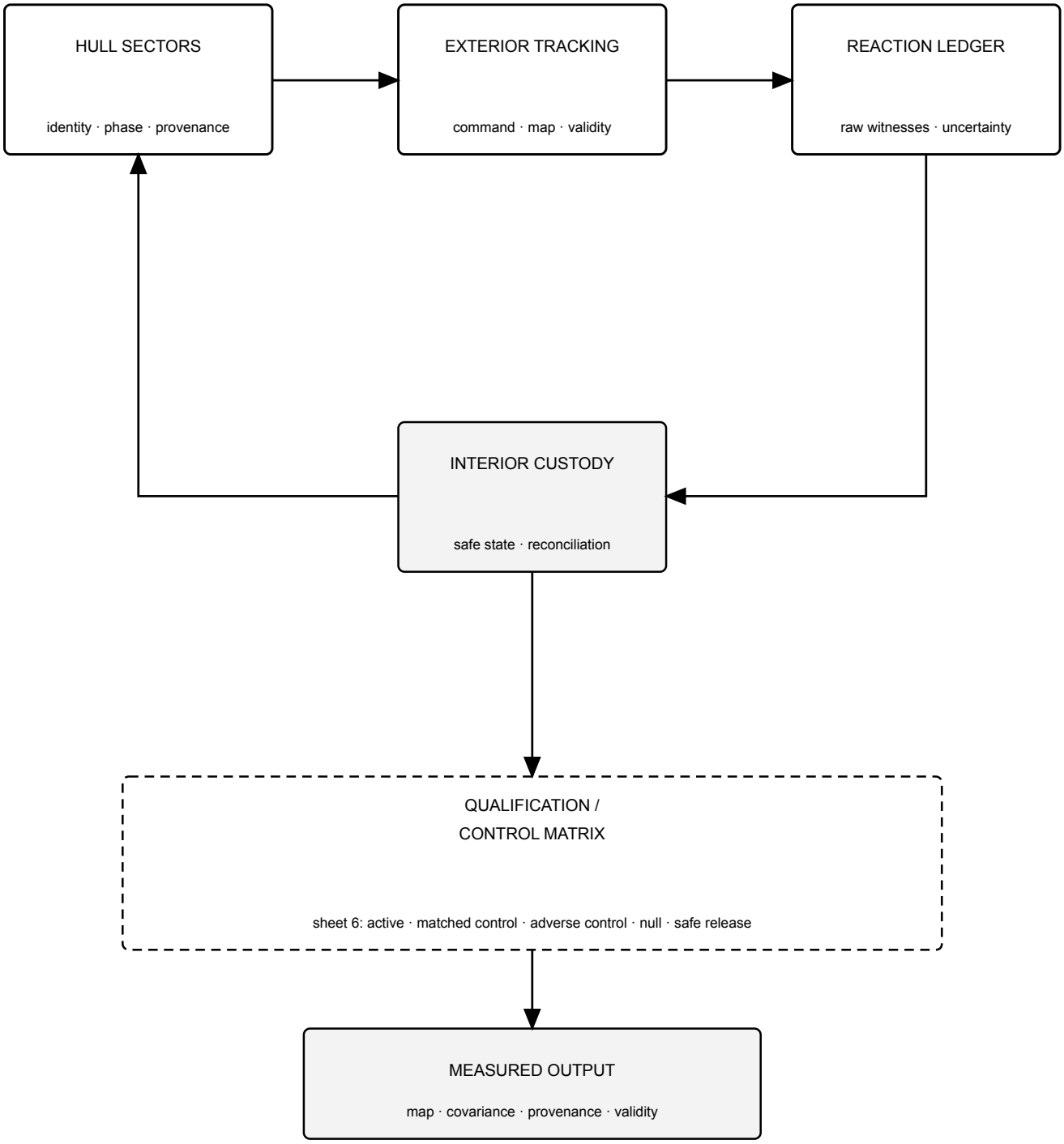


FIG. 6
Suspended reference test article and independently instrumented source/reaction frame.

Free-flight or free-fall qualification range with independent tracking and environmental witnesses

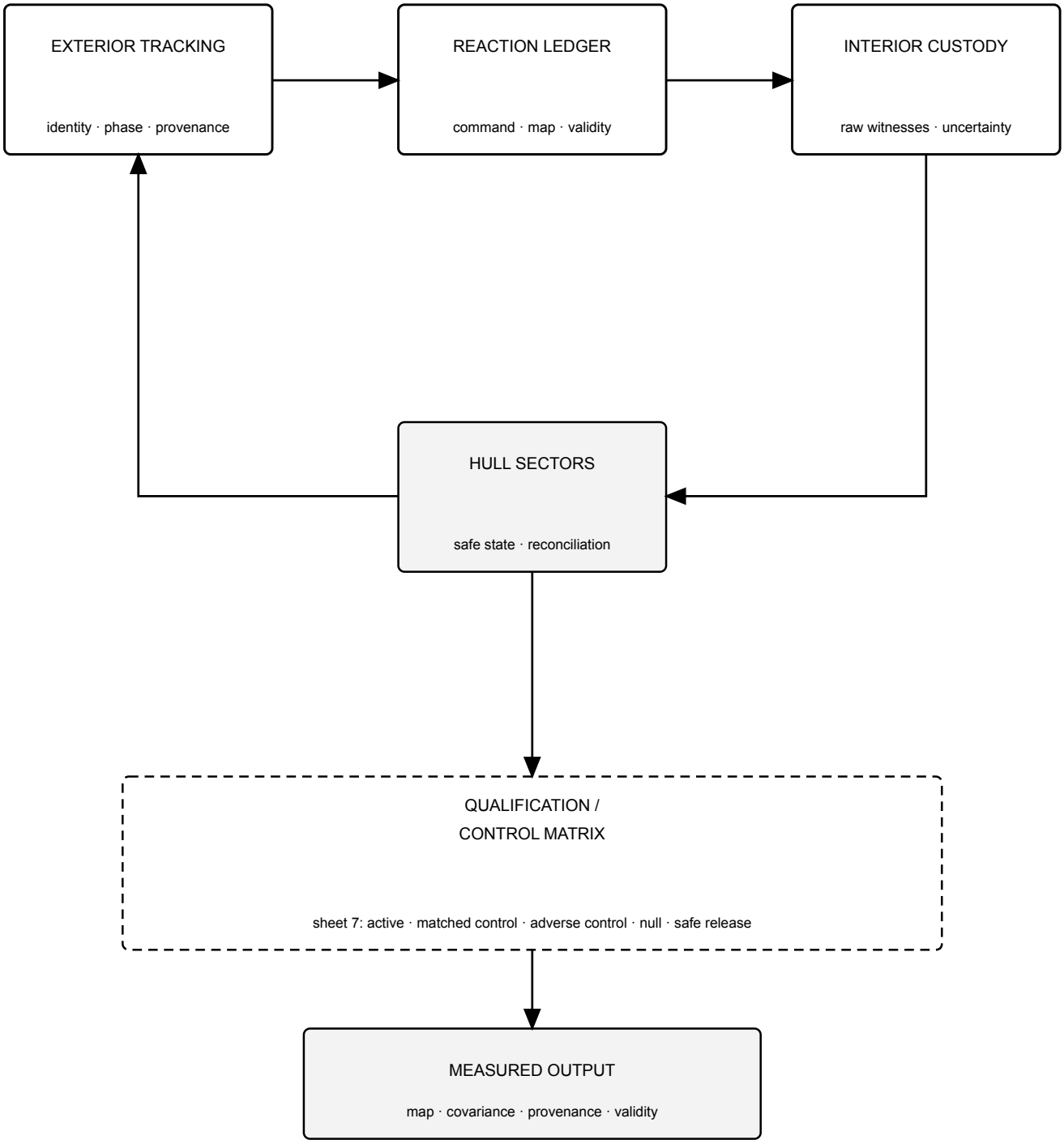


FIG. 7
Free-flight or free-fall qualification range with independent tracking and environmental witnesses.

Randomized control matrix and acceptance gates L1–L10

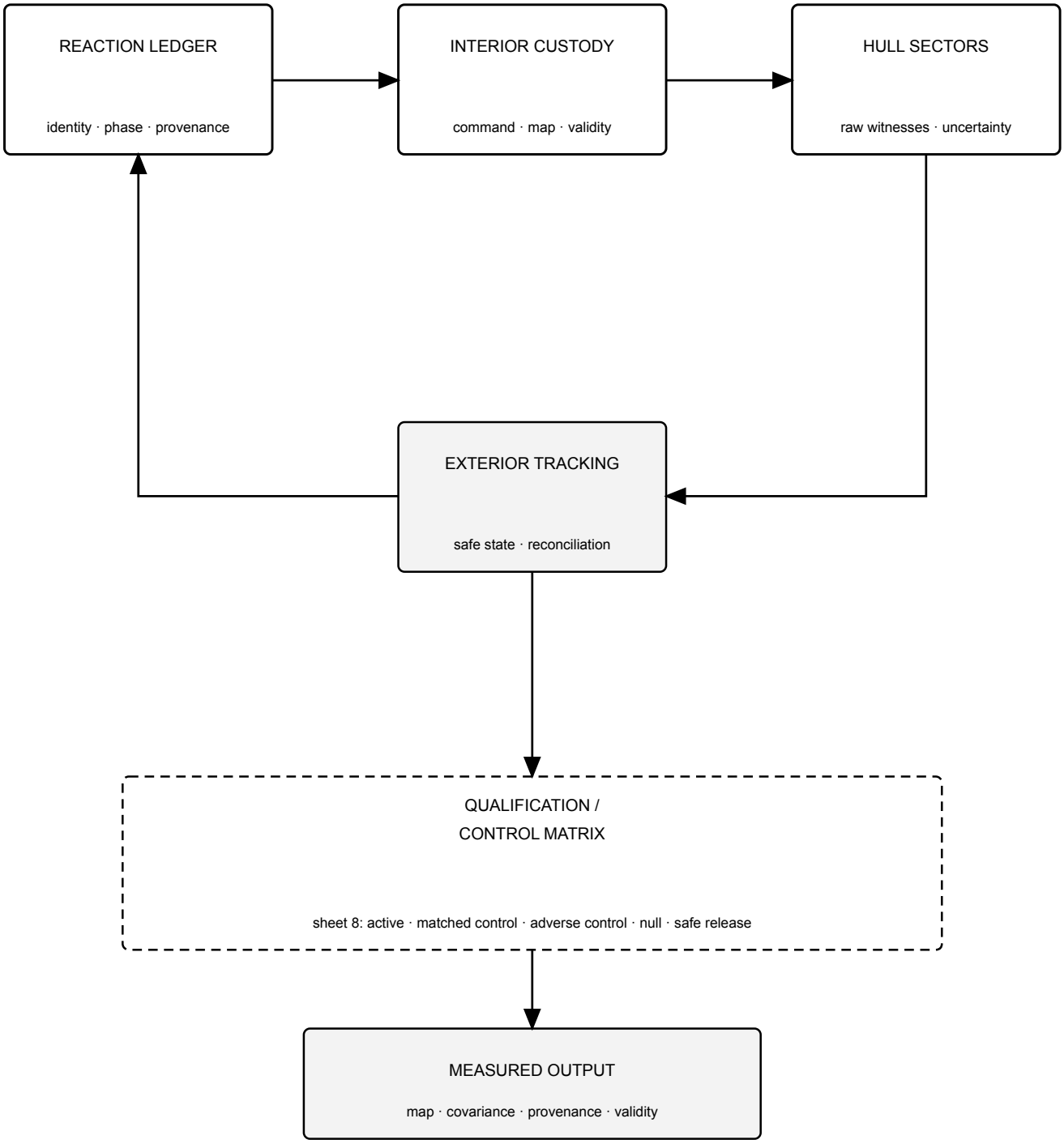


FIG. 8
Randomized control matrix and acceptance gates L1–L10.

Empirical translation-response map, permissible interpolation region, and array allocation

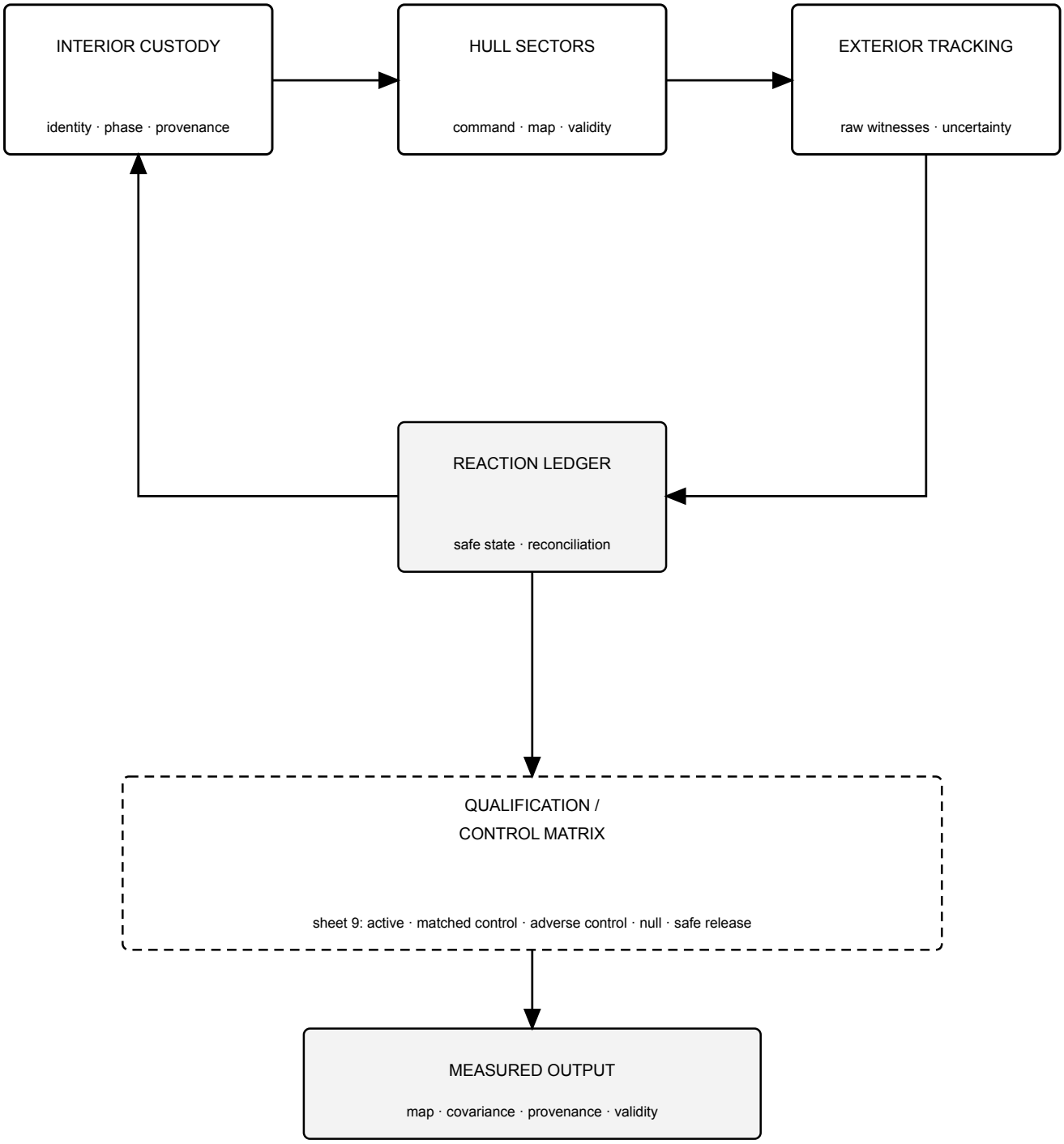


FIG. 9
Empirical translation-response map, permissible interpolation region, and array allocation.

Prophetic vehicle using a plural-sector hull, interior custody controller, and graceful-degradation modes

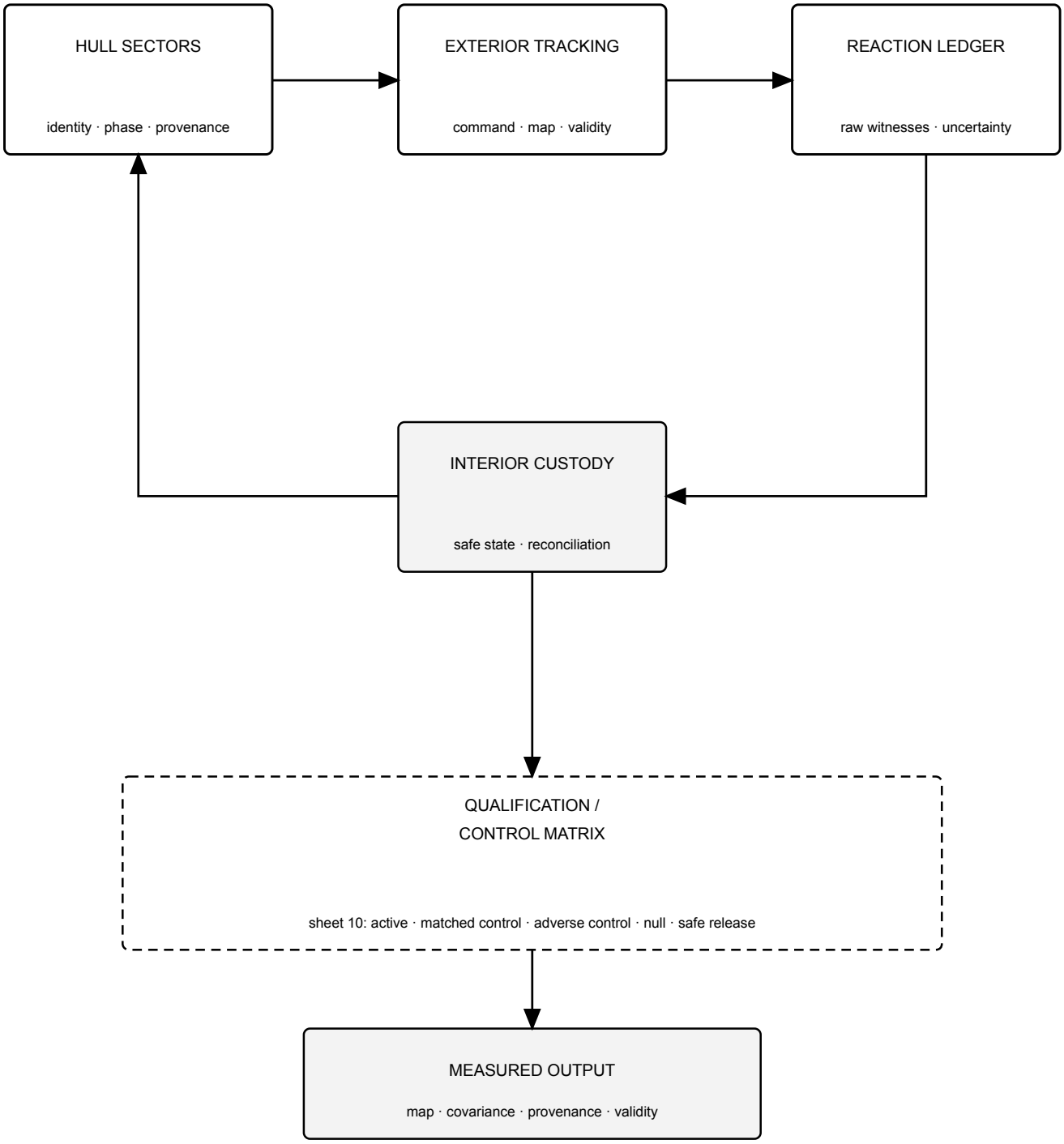


FIG. 10
Prophetic vehicle using a plural-sector hull, interior custody controller, and graceful-degradation modes.

Qualification-to-deployment, provenance binding, validity states, and recalibration

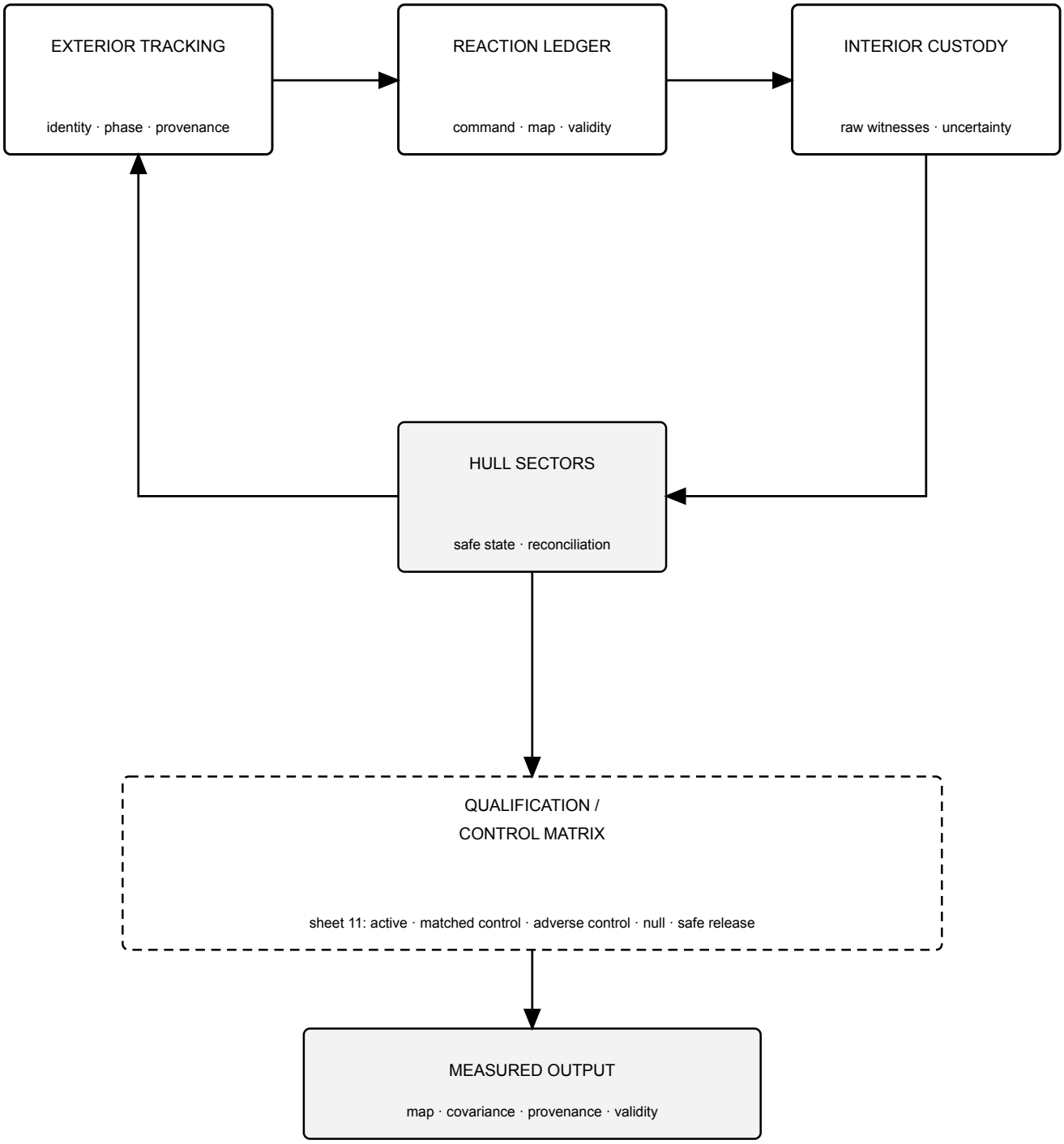


FIG. 11
Qualification-to-deployment, provenance binding, validity states, and recalibration.